

finetuning_1lm.ipynb - Colab

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finetuning_1lm.ipynb

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Commands Code Text Run all

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```
!pip install peft
```

```
Requirement already satisfied: peft in /usr/local/lib/python3.12/dist-packages (0.18.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from peft) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from peft) (25.0)
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from peft) (5.9.5)
Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages (from peft) (6.0.3)
Requirement already satisfied: torch>=1.13.0 in /usr/local/lib/python3.12/dist-packages (from peft) (2.9.0+cu126)
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (from peft) (4.57.1)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from peft) (4.67.1)
Requirement already satisfied: accelerate>=0.21.0 in /usr/local/lib/python3.12/dist-packages (from peft) (1.11.0)
Requirement already satisfied: safetensors in /usr/local/lib/python3.12/dist-packages (from peft) (0.7.0)
Requirement already satisfied: huggingface_hub>=0.25.0 in /usr/local/lib/python3.12/dist-packages (from peft) (0.36.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from huggingface_hub>=0.25.0->peft) (3.20.0)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub>=0.25.0->peft) (2025.3.0)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from huggingface_hub>=0.25.0->peft) (2.32.4)
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub>=0.25.0->peft) (4.15.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub>=0.25.0->peft) (1.2.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch>=1.13.0->peft) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.13.0->peft) (1.13.3)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.13.0->peft) (3.5)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.13.0->peft) (3.1.6)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch>=1.13.0->peft) (12.6.77)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch>=1.13.0->peft) (12.6.77)
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.6.80 in /usr/local/lib/python3.12/dist-packages (from torch>=1.13.0->peft) (12.6.80)
Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in /usr/local/lib/python3.12/dist-packages (from torch>=1.13.0->peft) (9.10.2.21)
Requirement already satisfied: nvidia-cublas-cu12==12.6.4.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.13.0->peft) (12.6.4.1)
Requirement already satisfied: nvidia-cufft-cu12==11.3.0.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.13.0->peft) (11.3.0.4)
Requirement already satisfied: nvidia-curand-cu12==10.3.7.77 in /usr/local/lib/python3.12/dist-packages (from torch>=1.13.0->peft) (10.3.7.77)
Requirement already satisfied: nvidia-cusolver-cu12==11.7.1.2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.13.0->peft) (11.7.1.2)
Requirement already satisfied: nvidia-cusparse-cu12==12.5.4.2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.13.0->peft) (12.5.4.2)
```

Variables

Terminal

2:25 PM T4 (Python 3)

finetuning_llm.ipynb - Colab

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finetuning_llm.ipynb

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RAM Disk

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[2] ✓ 2m

```
from transformers import AutoModelForSeq2SeqLM
from peft import get_peft_config, get_peft_model, LoraConfig, TaskType
model_name_or_path = "bigscience/mt0-large"
tokenizer_name_or_path = "bigscience/mt0-large"

peft_config = LoraConfig(
    task_type=TaskType.SEQ_2_SEQ_LM, inference_mode=False, r=8, lora_alpha=32, lora_dropout=0.1
)

model = AutoModelForSeq2SeqLM.from_pretrained(model_name_or_path)
model = get_peft_model(model, peft_config)
model.print_trainable_parameters()
"trainable params: 2359296 || all params: 1231940608 || trainable%: 0.19151053100118282"
```

⋮

... /usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF\_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/tokens>), set it as secret in your Google Colab and restart you
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
config.json: 100% ██████████ 800/800 [00:00<00:00, 34.0kB/s]
model.safetensors: 100% ██████████ 4.92G/4.92G [01:14<00:00, 94.3MB/s]
trainable params: 2,359,296 || all params: 1,231,940,608 || trainable%: 0.1915
'trainable params: 2359296 || all params: 1231940608 || trainable%: 0.19151053100118282'

[6]

from peft import AutoPeftModelForSeq2SeqLM

Variables

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T4 (Python 3)

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finetuning_llm.ipynb

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```
from peft import AutoPeftModelForCausalLM
from transformers import AutoTokenizer
import torch

model = AutoPeftModelForCausalLM.from_pretrained("ybelkada/opt-350m-lora").to("cuda")
tokenizer = AutoTokenizer.from_pretrained("facebook/opt-350m")

model.eval()
inputs = tokenizer("how is software jobs in future", return_tensors="pt")

outputs = model.generate(input_ids=inputs["input_ids"].to("cuda"), max_new_tokens=50)
print(tokenizer.batch_decode(outputs, skip_special_tokens=True)[0])

"Preheat the oven to 350 degrees and place the cookie dough in the center of the oven. In a large bowl, combine the flour, baking powder, baking soda, salt, and cinnamon. In
```

how is software jobs in future?

I am looking for a job in the software industry. I am a software developer and I have been working in the software industry for the past 3 years. I have been working in the s
'Preheat the oven to 350 degrees and place the cookie dough in the center of the oven. In a large bowl, combine the flour, baking powder, baking soda, salt, and cinnamon. In
a separate bowl, combine the egg yolks, sugar, and vanilla.'

Double-click (or enter) to edit

[] Start coding or generate with AI.

Variables Terminal

2:25 PM T4 (Python 3)